CYTECH ACADEMY

Digital Forensics & Incident Response

Complete Study Notes - beginner friendly, exam and lab ready.

(c) CyTech. For enrolled students. Unauthorized attack is illegal - practice only in labs.

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Forensic Fundamentals

Goal: preserve and analyze evidence so findings would stand in court. Rules: work only on forensic images (bit-by-bit copies), hash everything (MD5/SHA256) before and after, document every action with timestamps (chain of custody), never boot or modify the original.

Order of volatility: collect most-volatile first - RAM, network connections, running processes, then disk, then logs. Pulling the plug destroys RAM evidence; photograph the screen first.

Disk Forensics

Acquire with dd/FTK Imager (dd if=/dev/sdb of=image.dd bs=4M status=progress; sha256sum image.dd). Analyze in Autopsy: recover deleted files (filesystems often only remove the index entry), timeline file activity, keyword search, carve files from unallocated space with foremost/scalpel.

Windows artifacts worth gold: Prefetch (program execution), registry hives (user activity, USB devices via USBSTOR), $MFT, browser history, event logs (Event ID 4624 logon, 4688 process creation). Linux: /var/log/auth.log, bash_history, cron, journalctl.

Memory Forensics

Volatility 3 on a RAM dump: windows.pslist (processes), windows.netscan (connections), windows.cmdline, windows.malfind (injected code - classic malware hiding spot), windows.handles. Memory reveals what disk hides: unpacked malware, keys, in-memory-only implants.

Incident Response Lifecycle

NIST cycle: Preparation -> Detection & Analysis -> Containment -> Eradication -> Recovery -> Lessons Learned. Detection sources: SIEM alerts (Wazuh), EDR, IDS, user reports. During analysis, classify severity and scope: which hosts, which accounts, what data.

Containment: isolate the host from the network (not power off if you need RAM), reset compromised credentials, block IOCs at firewall/DNS. Eradication: remove malware, close the entry vector (patch!). Recovery: restore from known-clean backups, monitor aggressively for re-infection.

Building an IR-Ready Environment

Centralize logs off-box (attackers wipe local logs), enable PowerShell/process command-line logging, keep offline backups (3-2-1 rule), maintain an asset inventory and an IR runbook with contacts. Practice with tabletop scenarios and platforms like LetsDefend/CyberDefenders.